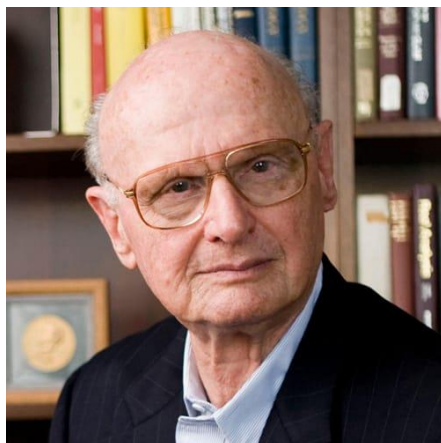


Portfolio Management

(1. Modern Portfolio Theory (MPT))

The genesis



(Harry Markowitz)

- ▶ Before Markowitz, portfolio construction was focused on **expected return** with minimal or no regards to risk.
- ▶ The idea of '*do not put all your eggs in 1 basket*' was prevalent but there was no quantification of risk.
- ▶ Harry Markowitz formalized the notion of risk as Variance or Standard deviation of the asset returns. This opened the flood gates of quantitative techniques and optimization surrounding a portfolio's risk-return characteristics.
- ▶ Harry showed that risk reduction is possible using diversification through correlation between asset returns.
- ▶ Harry's protégé 'William Sharpe' proposed a linear one factor model as an equilibrium model which became famous as CAPM (Capital Asset Pricing Model).
- ▶ Both Harry and Sharpe were awarded Nobel Prize in 1990 for their contribution to financial economics.



The Set up

- ▶ We have $N \geq 2$ assets to invest in
- ▶ We want to invest our initial wealth \$ V into these assets
- ▶ For now, assume buying and shorting both are allowed
- ▶ Our goal is to find the best investment taking both risk and return into the consideration
- ▶ Fractional investing is possible
- ▶ Markets are frictionless, .e.g. no transaction costs, no collaterals or margins
- ▶ The assets are encompassing everything that is tradeable on the planet earth i.e., financial assets (bonds, stocks), real assets (commodities, real estate) or even intangible assets like human capital

From Asset Return to Portfolio Return

- ▶ Portfolio returns (arithmetic) is linear transformation of the assets returns

Assets	1	2
Price	S_1	S_2
Units	n_1	n_2
Value	$V_1 = n_1 S_1$	$V_2 = n_2 S_2$
Weights	$w_1 = \frac{V_1}{V_1 + V_2}$	$w_2 = \frac{V_2}{V_1 + V_2}$

Assets	1	2
Price	$S_1 + \Delta S_1$	$S_2 + \Delta S_2$
Units	n_1	n_2
Returns	$R_1 = \frac{\Delta S_1}{S_1}$	$R_2 = \frac{\Delta S_2}{S_2}$
Mean Return	$\mathbb{E}[R_1] = \mu_1$	$\mathbb{E}[R_2] = \mu_2$
SD	$\mathbb{V}[R_1] = \sigma_1^2$	$\mathbb{V}[R_2] = \sigma_2^2$

$t = 0$

$t = 1$

▶ Portfolio return :

$$R_P = \frac{n_1(\Delta S_1) + n_2(\Delta S_2)}{V_1 + V_2} = \frac{n_1 S_1 (\Delta S_1 / S_1) + n_2 S_2 (\Delta S_2 / S_2)}{V_1 + V_2} = w_1 R_1 + w_2 R_2$$



Basic Algebra from Statistics

► There are two fundamental equations we will be needing for portfolio risk and return.

- Expectation

Expectation is a linear operator. Expectation of a linear combination is the linear combination of expectation.

$$\mathbb{E}[\alpha X \pm \beta Y] = \alpha \mathbb{E}[X] \pm \beta \mathbb{E}[Y]$$

$$\mathbb{E}[\mathbf{w}^T \mathbf{X}] = \mathbf{w}^T \mathbb{E}[\mathbf{X}] \quad (\text{In matrix form})$$

- Variance

Variance is not a linear operator.

$$\mathbb{V}[\alpha X \pm \beta Y] = \alpha^2 \mathbb{V}[X] + \beta^2 \mathbb{V}[Y] \pm 2\alpha\beta \text{Cov}(X, Y)$$

$$\mathbb{V}[\mathbf{w}^T \mathbf{X}] = \mathbf{w}^T \mathbf{\Omega} \mathbf{w} \quad (\text{In matrix form, } \mathbf{\Omega} \text{ is the Cov matrix})$$



Portfolio Return moments

- In Markowitz world, the investor has a **mean-variance utility function**; meaning the only two moments of the portfolio that matter are the expected return and variance. In other words, we need to find out two quantities $\mathbb{E}[R_p]$ and $\mathbb{V}[R_p]$. Algebra from statistics comes to the rescue, thanks to the linear nature of portfolio return

(In matrix form)

$$R_p = w_1 R_1 + w_2 R_2$$

$$\mathbb{E}[R_p] = w_1 \mathbb{E}[R_1] + w_2 \mathbb{E}[R_2] = w_1 \mu_1 + w_2 \mu_2$$

$$\mathbb{V}[R_p] = w_1^2 \mathbb{V}[R_1] + w_2^2 \mathbb{V}[R_2] + 2w_1 w_2 \text{Cov}(R_1, R_2)$$

$$\mathbb{V}[R_p] = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$$

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \quad \mathbb{E}[\mathbf{R}] = \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} R_1 \\ R_2 \end{pmatrix} \quad \boldsymbol{\Omega} = \begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix}$$

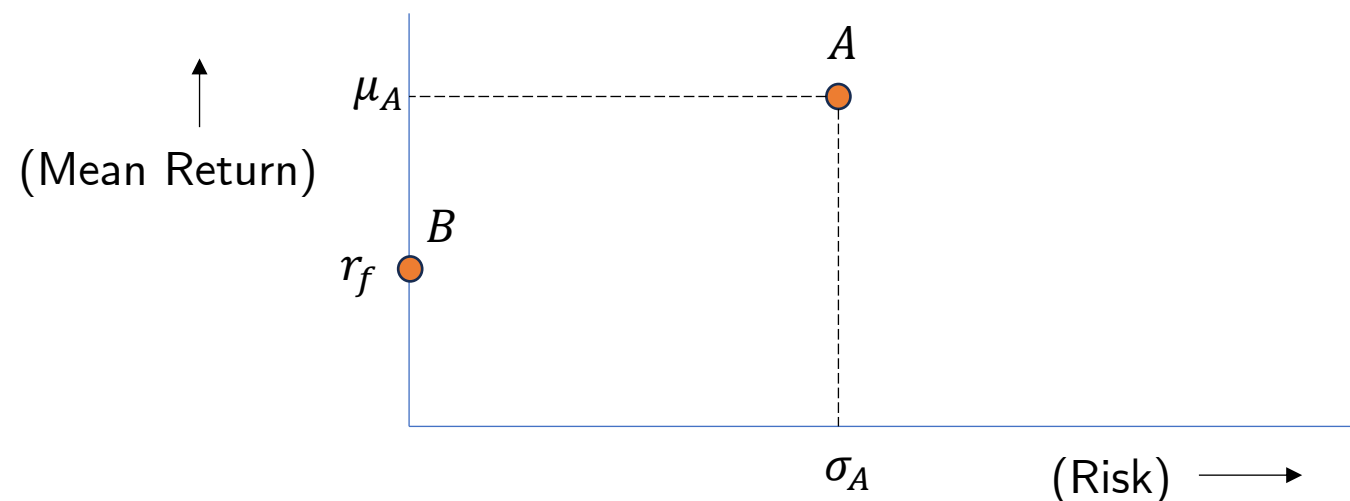
$$R_p = \mathbf{w}^T \mathbf{R}$$

$$\mathbb{E}[R_p] = \mathbf{w}^T \mathbb{E}[\mathbf{R}] = \mathbf{w}^T \boldsymbol{\mu}$$

$$\mathbb{V}[R_p] = \mathbf{w}^T \boldsymbol{\Omega} \mathbf{w}$$

I. One risky and One Risk-free asset

- ▶ Consider a portfolio of two assets : 1 risky asset (A) and 1 risk-free asset (B)
- ▶ The risk-free asset has the following properties
 - ▶ constant return = r_f ,
 - ▶ has volatility = 0 and
 - ▶ correlation = 0 with the risky asset
- ▶ Let the weights invested in A and B be w_A and w_B respectively. Of course, $w_A + w_B = 1$



I. One risky and One Risk-free asset

► The Mean return and the risk of the portfolio parametrized by w_A is given as

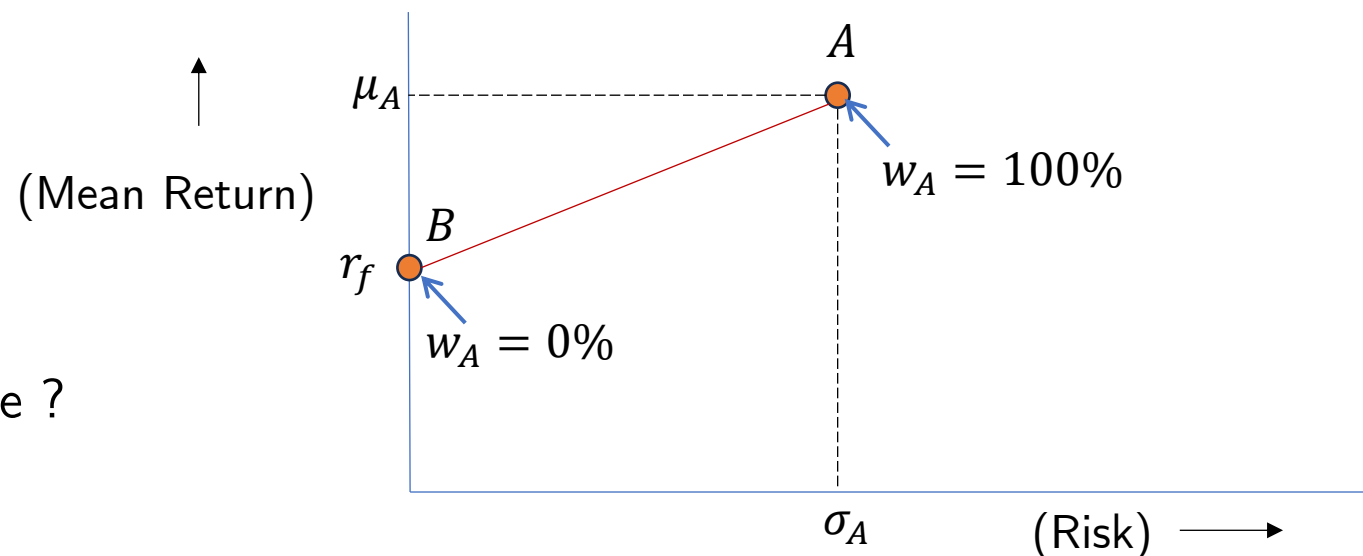
$$\mathbb{E}[R_P] = w_1\mu_1 + w_2\mu_2 = w_A\mu_A + (1 - w_A)r_f = r_f + w_A(\mu_A - r_f)$$

$$\mathbb{V}[R_P] = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2 = w_A^2\sigma_A^2 + (1 - w_A)^2(0)^2 + 2w_A(1 - w_A)0 \cdot \sigma_A \cdot 0 = w_A^2\sigma_A^2$$

$$\mu_P = r_f + w_A(\mu_A - r_f)$$

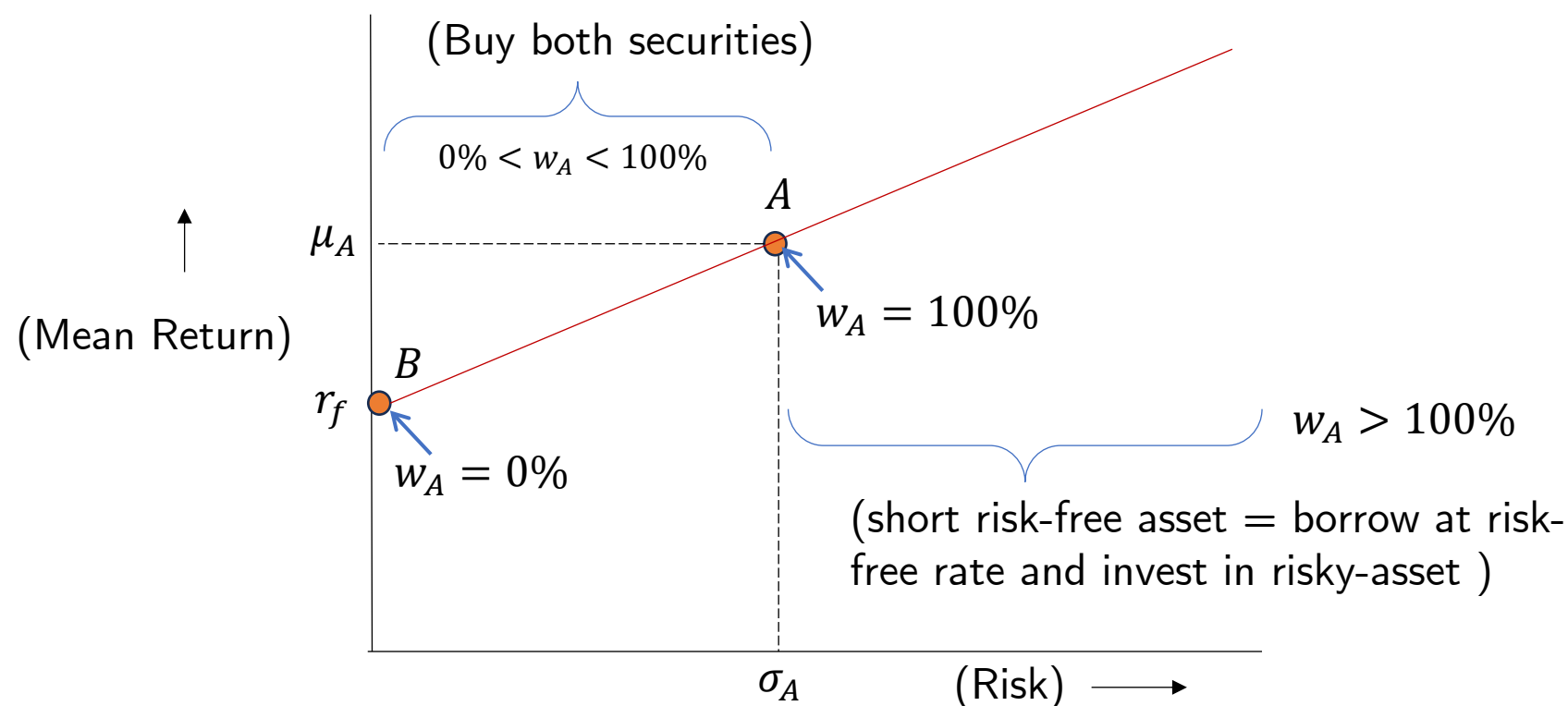
$$\sigma_P = w_A\sigma_A$$

► What is the intercept and slope of the line ?



I. One risky and One Risk-free asset

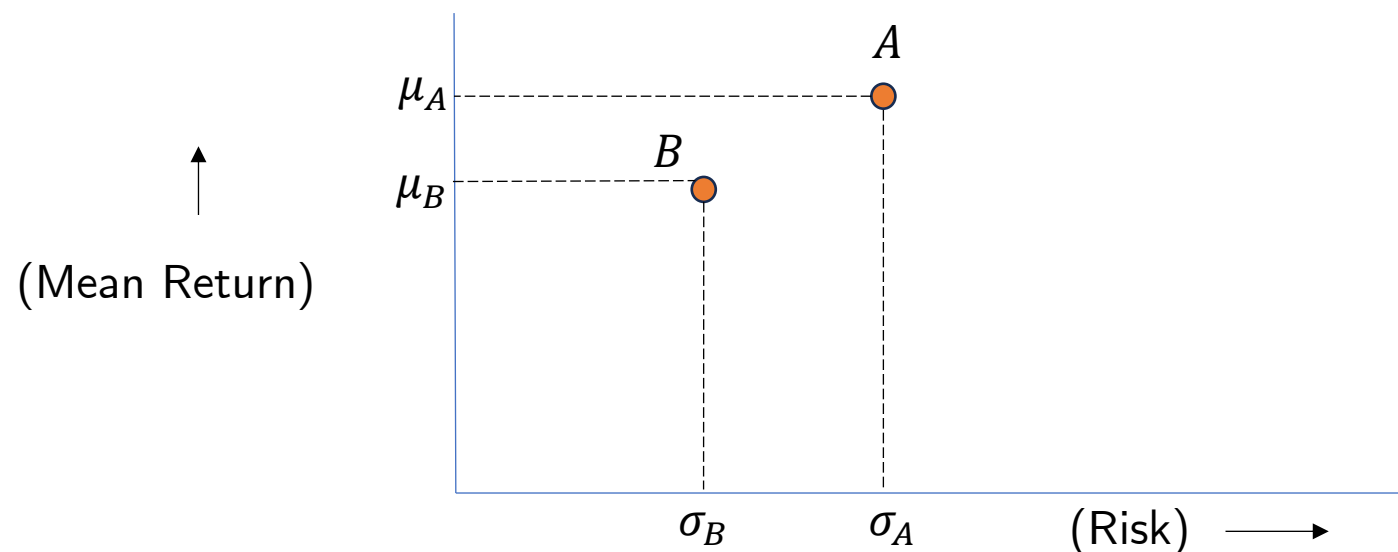
- We can move outside of the extremities if we allow short selling.



- What if we short the risky asset and invest on risk-free asset ?

II. Two risky assets

- ▶ Consider a portfolio of two risky assets : (A) and (B)
- ▶ Let the weights invested in A and B be w_A and w_B respectively. Of course, $w_A + w_B = 1$
- ▶ What is the risk-return profile of the portfolio ?



II. Two risky assets

- ▶ The mean-return and the risk of the portfolio are given as

$$\mathbb{E}[R_P] = w_1\mu_1 + w_2\mu_2 = w_A\mu_A + (1 - w_A)\mu_B = \mu_B + w_A(\mu_A - \mu_B)$$

$$\mathbb{V}[R_P] = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2 = w_A^2\sigma_A^2 + (1 - w_A)^2\sigma_B^2 + 2w_A(1 - w_A)\rho\sigma_A\sigma_B$$

$$\mu_P = \mu_B + w_A(\mu_A - \mu_B)$$

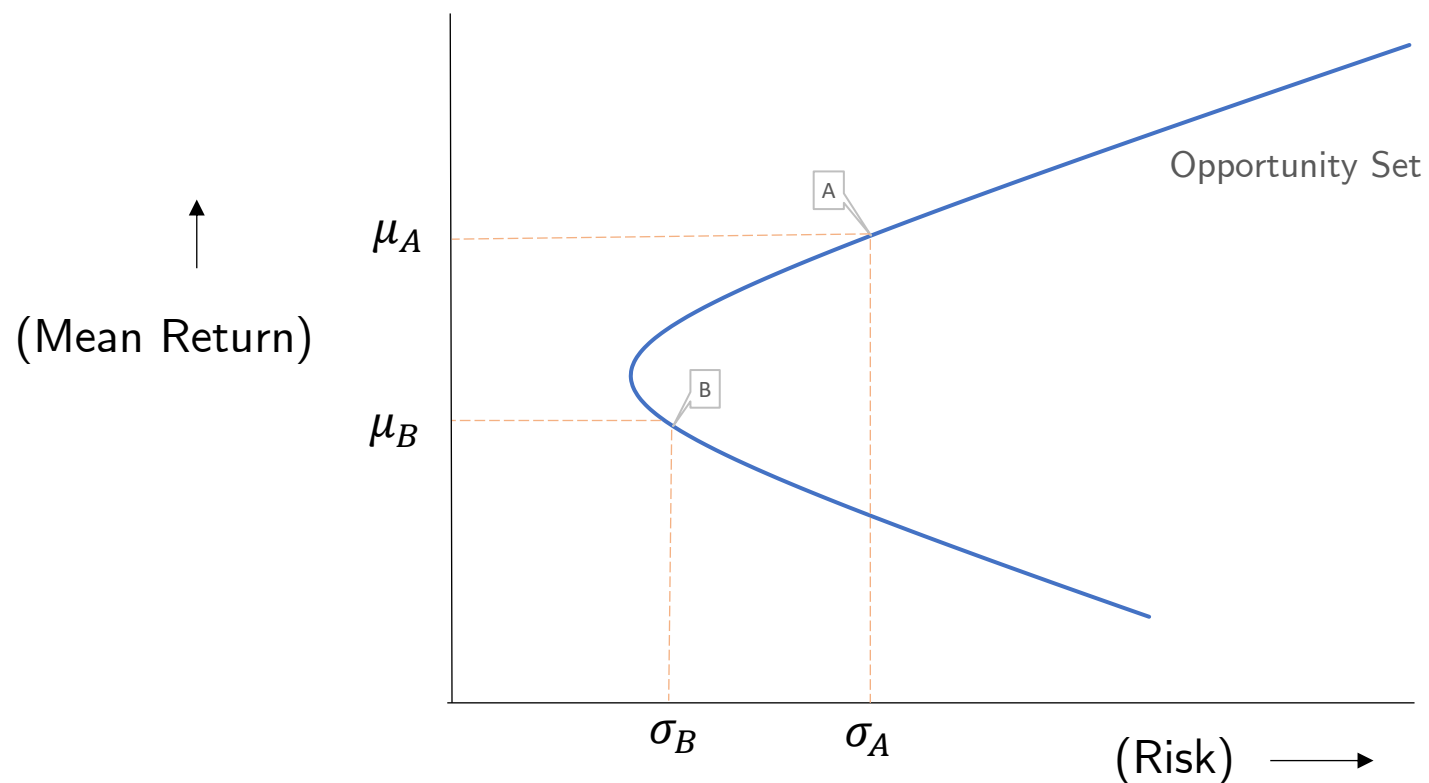
$$\sigma_P = \sqrt{w_A^2\sigma_A^2 + (1 - w_A)^2\sigma_B^2 + 2w_A(1 - w_A)\rho\sigma_A\sigma_B}$$

- ▶ Also note that the volatility of the portfolio returns also depends on the correlation $-1 \leq \rho \leq 1$. One gets maximum diversification with $\rho = -1$ and no diversification when $\rho = +1$
- ▶ The volatility is bounded between

$$|w_A\sigma_A - w_B\sigma_B| \leq \sigma_P \leq |w_A\sigma_A + w_B\sigma_B|$$

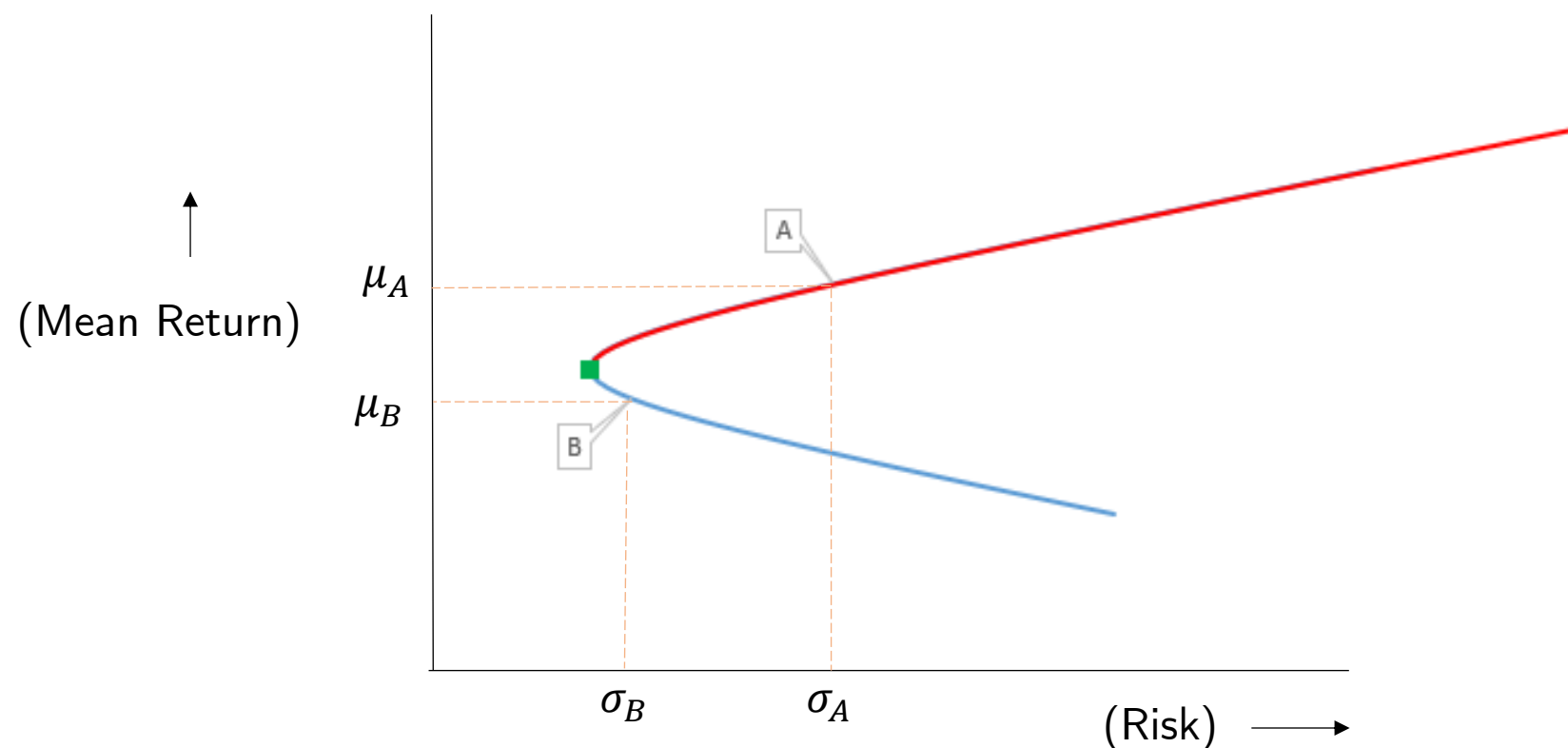
II. Two risky assets

- ▶ The profile we get is a hyperbola. It's called a the "Opportunity Set". Investor can choose to be anywhere on the opportunity set.
- ▶ Are all points on the curve efficient ?



II. Two risky assets – Efficient Frontier

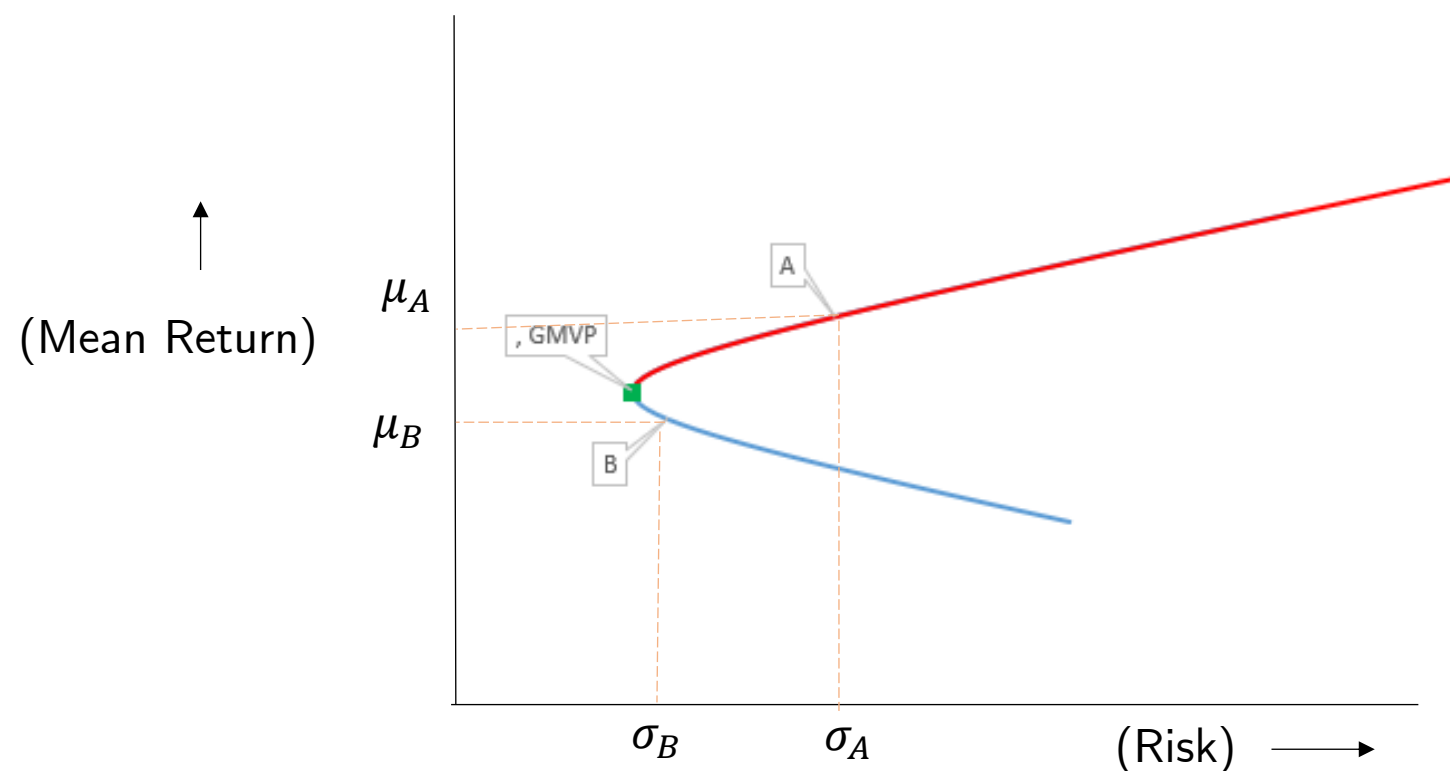
- **Efficient Frontier** is defined as the part of the opportunity set which gives the best risk-return combination.



II. Two risky assets – GMVP

► There exists a point which corresponds to the minimum variance. This is called as **Global Minimum Variance Portfolio (GMVP)**

► The weight of A that corresponds to GMVP is given as $w_G^A = \frac{\sigma_B(\sigma_B - \rho\sigma_A)}{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}$

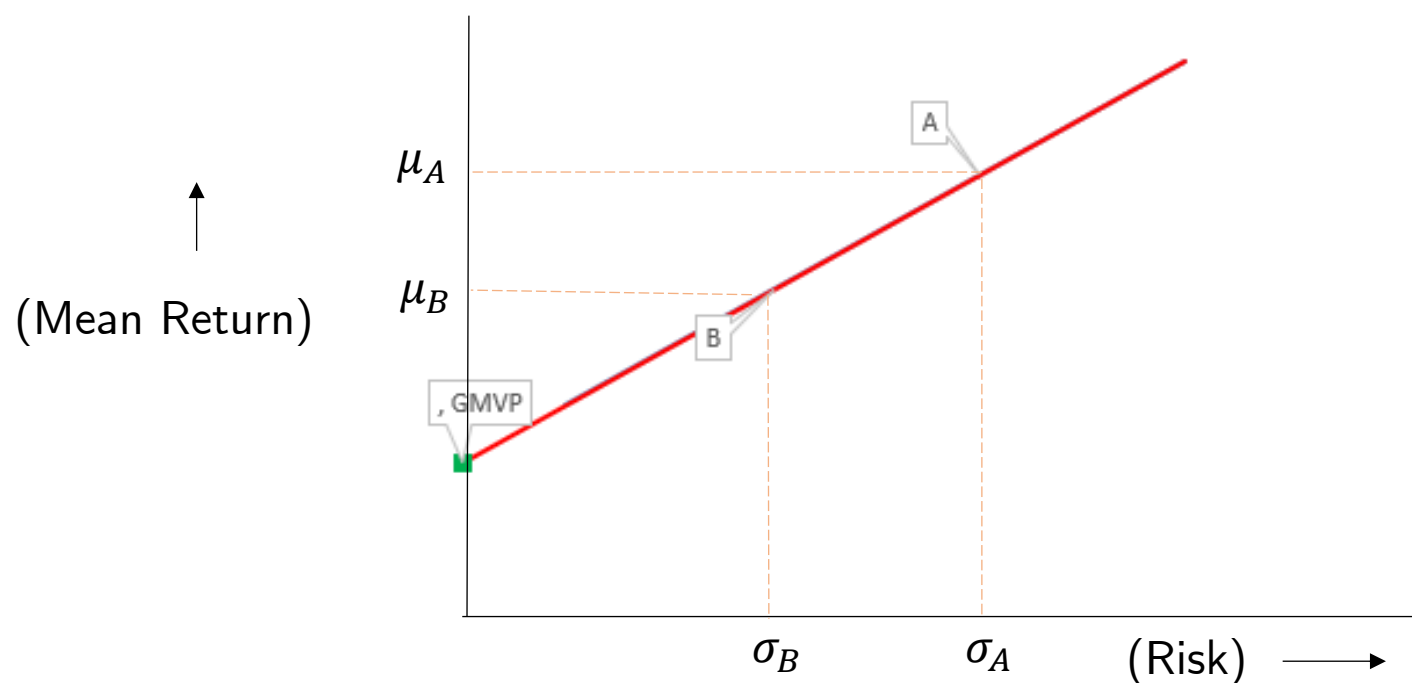


II. Two risky assets : Case $\rho = 1$

- If assets are perfectly positively correlated, there is no diversification benefit. The equations for mean return and volatility of the portfolio take the following form. It's possible to get a zero-variance portfolio.

$$\mu_P = \mu_B + w_A(\mu_A - \mu_B)$$

$$\sigma_P = |w_A\sigma_A + w_B\sigma_B| = |\sigma_B + w_A(\sigma_A - \sigma_B)|$$

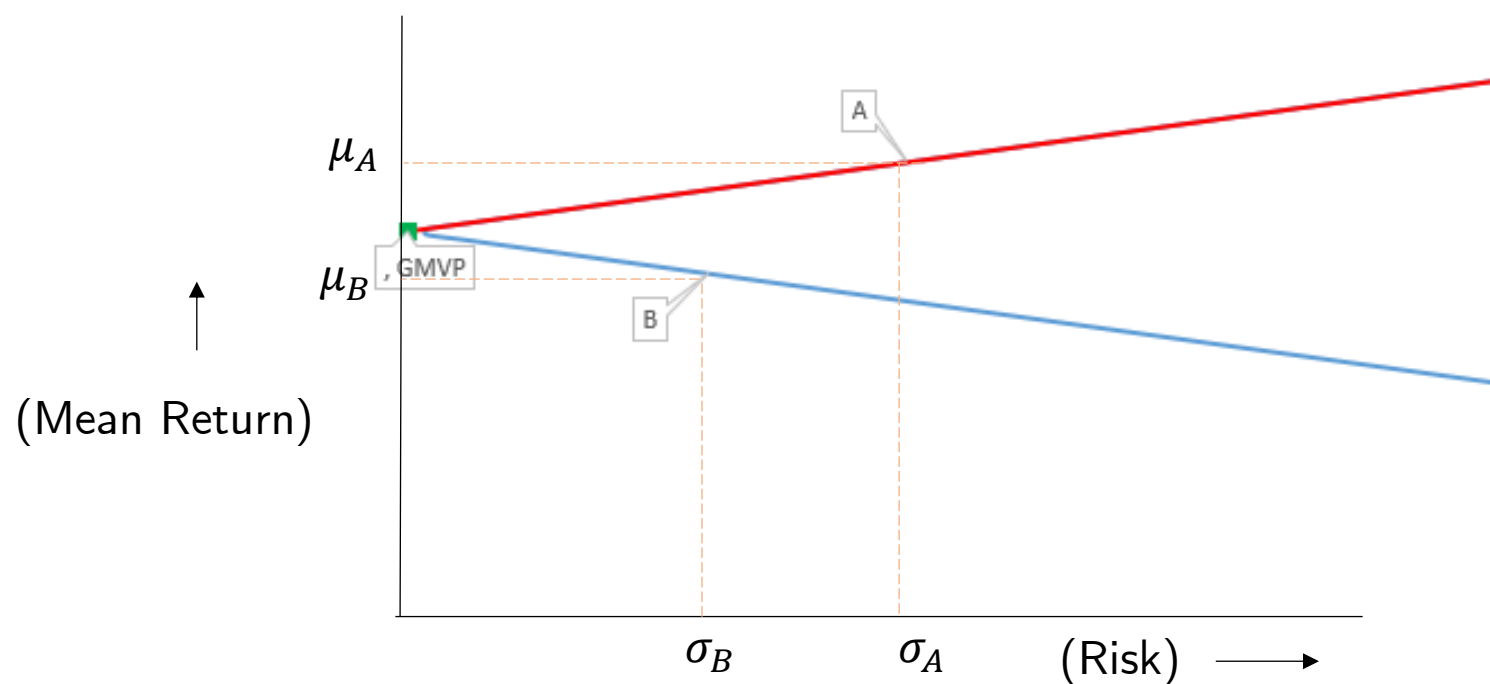


II. Two risky assets : Case $\rho = -1$

- If assets are perfectly negatively correlated, the diversification benefit is the highest. The equations for mean return and volatility of the portfolio take the following form. It's possible to get a zero-variance portfolio.

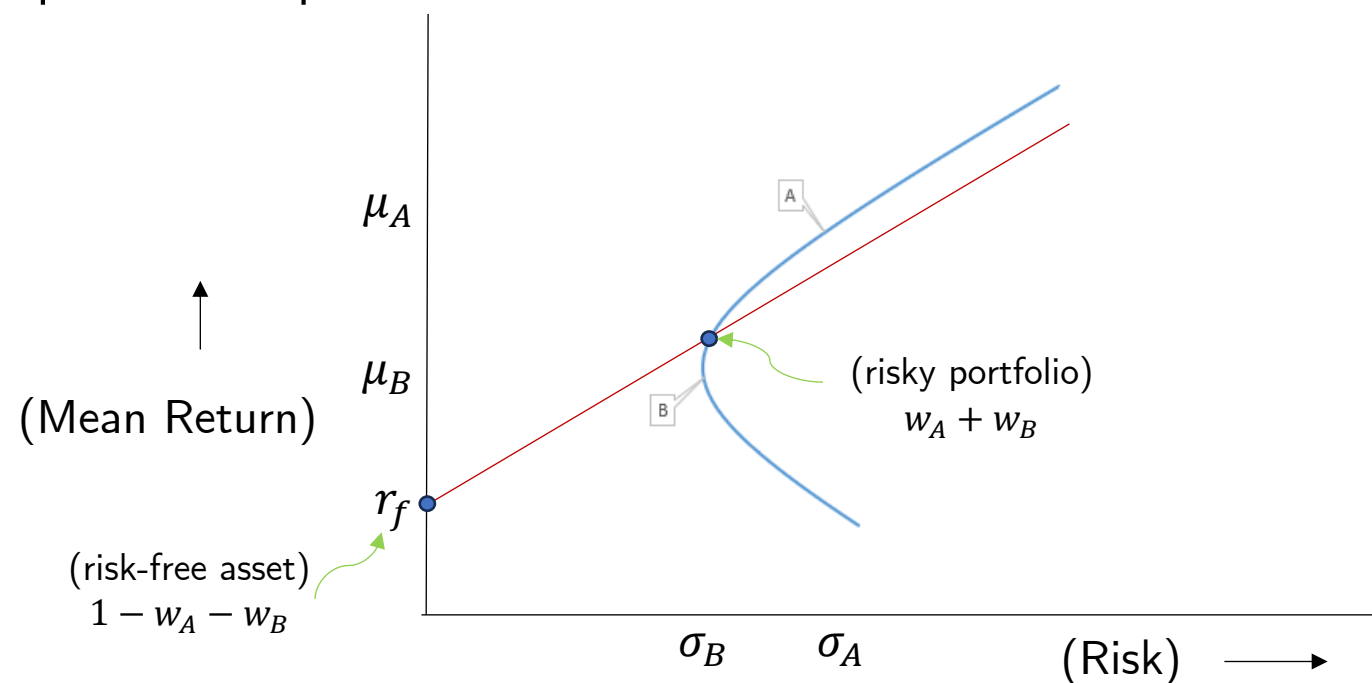
$$\mu_P = \mu_B + w_A(\mu_A - \mu_B)$$

$$\sigma_P = |w_A\sigma_A - w_B\sigma_B| = |-\sigma_B + w_A(\sigma_A + \sigma_B)|$$



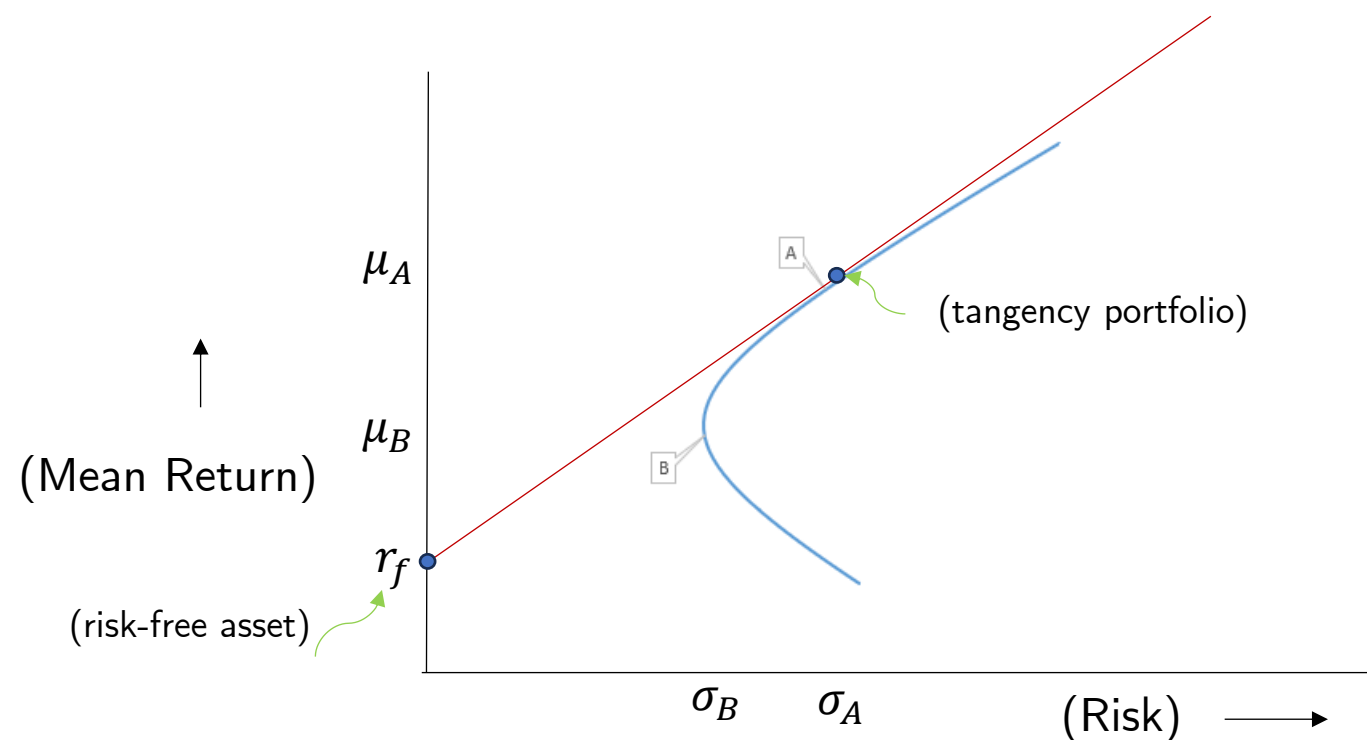
III. Reviving the Risk-Free Asset

- ▶ We can now consider the risk-free asset and the portfolio of risky assets; the latter can be thought of one risky asset. We have already seen that in such a case, we get a linear risk-return profile joining the characteristics of risk-free asset and risky portfolio.
- ▶ So basically, $w_A + w_B$ would be invested in the risky portfolio and $(1 - w_A - w_B)$ will be invested in the risk-free asset. But, changing w_A, w_B allows us to move along the hyperbola; in other words, we have a wide spectrum of possible risk-assets that we can mix with the risk-free asset.



III. The New Efficient Frontier

- ▶ With the risk-free asset and the risky portfolio, we can achieve higher efficiency by increasing the slope. The efficient frontier is the line is the tangent to the hyperbola. The point of intersection is the **tangency portfolio**. The new efficient frontier is called CML (Capital Market Line)





The Tangency Portfolio

- ▶ Let w_t be the weight of the tangency portfolio. The risk return characteristics of the total portfolio including the risk-free asset and the tangency portfolio are given as

$$\mu_P = w_t \mu_t + (1 - w_t) r_f = r_f + w_t (\mu_t - r_f)$$

$$\sigma_P = w_t \sigma_t$$

- ▶ The equation of CML is given as $\mu_P = r_f + \frac{\sigma_P (\mu_t - r_f)}{\sigma_t}$

$$\mu_P = r_f + S_t \sigma_P \quad \left(\text{where } S_t = \frac{(\mu_t - r_f)}{\sigma_t} \text{ is the slope} \right)$$

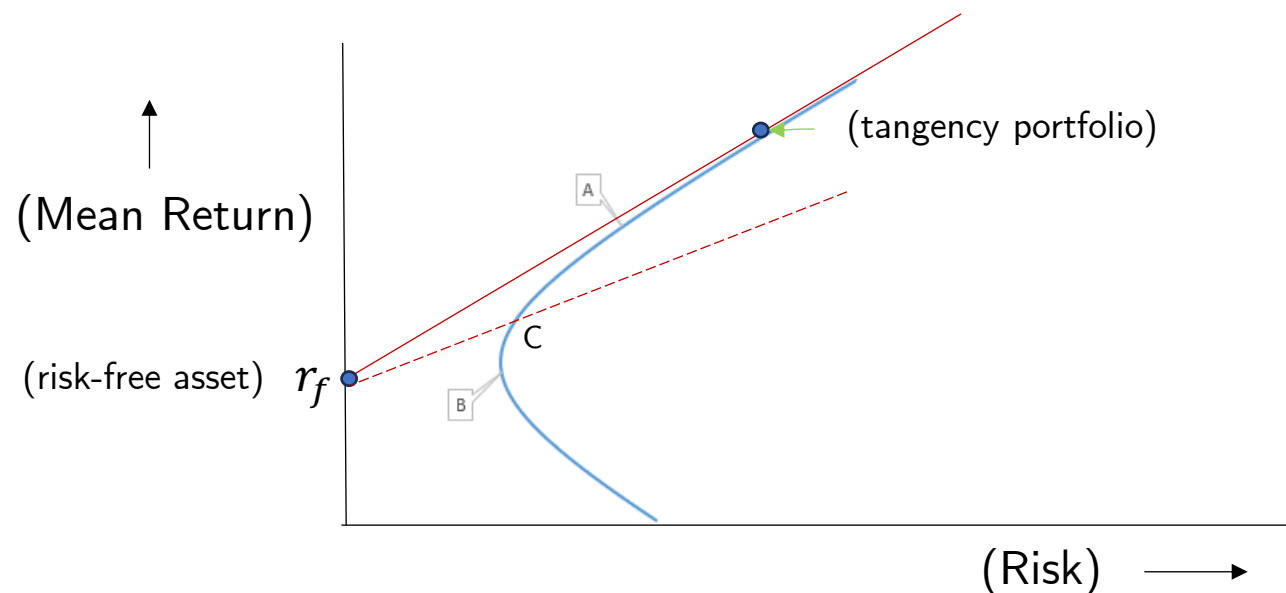
- ▶ S_t is the slope. It's a measure of increase in portfolio mean return for 1 unit change portfolio volatility. Clearly, larger the slope, more efficient the portfolio.

Sharpe Ratio

- ▶ For any risky portfolio (C), the Sharpe ratio is defined as

$$S_C = \frac{\mu_C - r_f}{\sigma_C}$$

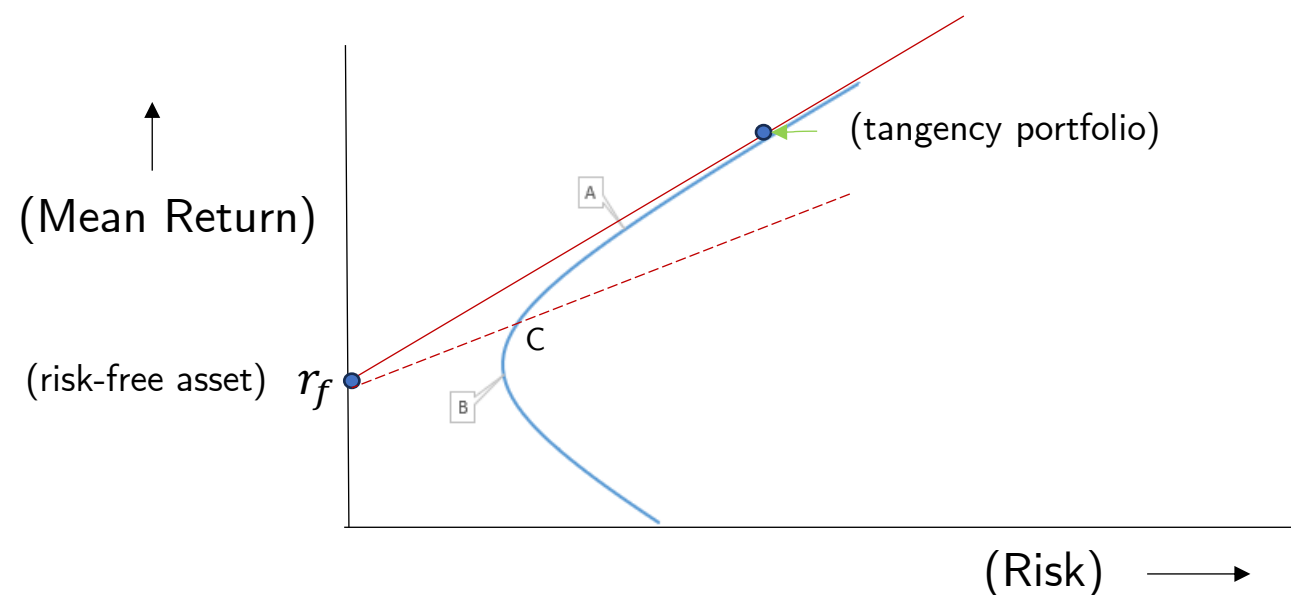
- ▶ The risk-return profile of the risk-free asset + any risky portfolio C , will be straight-line with the slope (S_C). This slope drives the efficiency of the portfolio. The tangency portfolio is the one with the highest Sharpe ratio and therefore lies on the efficient frontier.



Tangency Portfolio Allocation

- The tangency portfolio is fully invested in the risky assets A and B. The weight allocation to A is given as

$$w_A^t = \frac{\sigma_B \left((\mu_B - r_f) \rho \sigma_A - (\mu_A - r_f) \sigma_B \right)}{- (\mu_A - r_f) \sigma_B^2 - (\mu_B - r_f) \sigma_A^2 + (\mu_A + \mu_B - 2r_f) \rho \sigma_A \sigma_B}$$





More on Diversification

- ▶ Consider a homogeneous portfolio
 - ❑ N risky assets $i = 1, 2, \dots, N$
 - ❑ same expected return, e.g., $\mu_i = \mu$
 - ❑ same standard deviation of return, e.g., $\sigma_i = \sigma$
 - ❑ All pairwise correlations are same, e.g., $\rho_i = \rho$
- ▶ For convenience also assume we invest equally on all assets e.g., $w_i = w = \frac{1}{N}$
- ▶ The portfolio risk return characteristics are given as

$$\mu_P = \sum_{i=1}^N w_i \mu_i = \mu$$

(Invariant of N)

$$\sigma_P^2 = \left(\rho + \frac{1-\rho}{N} \right) \sigma^2$$

when $\rho = 0$, $\sigma_P^2 = \frac{\sigma^2}{N}$

- ▶ One can reduce risk by adding more securities. This is called diversification. However, In practice, there is a lower limit of risk that can be achieved in a market

Optimization with n assets

- ▶ Consider n risky assets fully characterized by their expected returns μ_i , standard deviation σ_i . Additionally, the correlation between asset i and j is given as ρ_{ij} . The weight of asset i is w_i
- ▶ Let's define some vectors and matrices or easy formulation

(Expected Return vector)

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix}$$

(Covariance Matrix)

$$\boldsymbol{\Omega} = \begin{pmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \cdots & \rho_{1n}\sigma_1\sigma_n \\ \rho_{21}\sigma_2\sigma_1 & \sigma_2^2 & \cdots & \rho_{2n}\sigma_2\sigma_n \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1}\sigma_n\sigma_1 & \cdots & \cdots & \sigma_n^2 \end{pmatrix}$$

(Weight vector)

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}$$

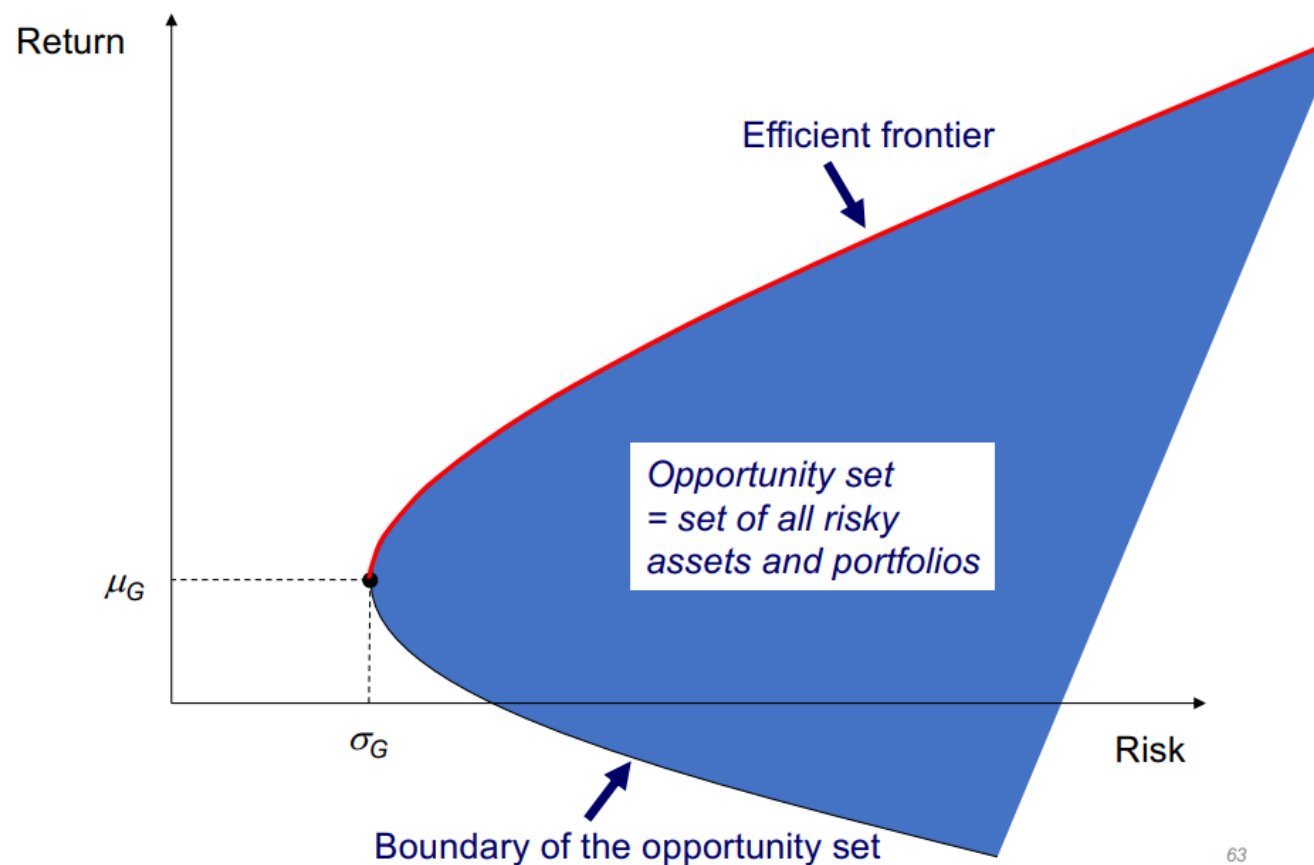
- ▶ Portfolio risk-return is given as

$$\mu_P = \mathbf{w}^T \boldsymbol{\mu}$$

$$\sigma_P^2 = \mathbf{w}^T \boldsymbol{\Omega} \mathbf{w}$$

Optimization with n assets

- ▶ With multiple assets, the opportunity set is the entire region bounded by the hyperbola.



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Constrained Optimization

► Optimization Case I : (Risk Constraint)

One approach would be to maximize mean return while keeping risk to a limit.

$$\max_{w_1, w_2, \dots, w_n} \mu_P$$

$$\text{Subject to: } \sigma_P^2 = \nu^2$$

$$\sum w_i = 1$$

► Optimization Case II : (Return Constraint)

$$\min_{w_1, w_2, \dots, w_n} \sigma_P^2$$

$$\text{Subject to: } \mu_P = m$$

$$\sum w_i = 1$$



Constrained Optimization

- ▶ Out of the two cases, Case II is more dominant. The reason is it's simpler for an investor to have return-based goals than risk-based goals. Let's use matrix notation to simplify our problem statement

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \mathbf{\Omega} \mathbf{w}$$

$$\text{Subject to: } \mathbf{w}^T \mathbf{\mu} = m$$

$$\mathbf{w}^T \mathbf{1} = 1$$

- ▶ Note that a factor of $\frac{1}{2}$ in the objective function is just a convention (mathematical convenience) and it does not affect the solution. Also, $\mathbf{1}$ is the vector of 1s
- ▶ The above constrained optimization problem can be converted to an unconstrained optimization problem using **Lagrangian**. Remember we can use Lagrangian multipliers with equality constraints.

Constrained Optimization

► Lagrangian formulation gives

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \frac{1}{2} \mathbf{w}^T \boldsymbol{\Omega} \mathbf{w} + \lambda(m - \mathbf{w}^T \boldsymbol{\mu}) + \gamma(1 - \mathbf{w}^T \mathbf{1})$$

Differentiating w.r.t $\mathbf{w}$ gives $\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \boldsymbol{\Omega} \mathbf{w} - \lambda \boldsymbol{\mu} - \gamma \mathbf{1} = \mathbf{0}$

The optimal allocation vector $\mathbf{w}^* = \boldsymbol{\Omega}^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1})$

Substituting $\mathbf{w}^*$ in the constraints will give the values for λ and γ

These are conveniently expressed in terms of three scalars (A, B, C)

$$\lambda = \frac{Am - B}{AC - B^2}$$

$$\gamma = \frac{C - Bm}{AC - B^2}$$

$$\text{where } \left\{ \begin{array}{l} A = \mathbf{1}^T \boldsymbol{\Omega}^{-1} \mathbf{1} \\ B = \boldsymbol{\mu}^T \boldsymbol{\Omega}^{-1} \mathbf{1} \\ C = \boldsymbol{\mu}^T \boldsymbol{\Omega}^{-1} \boldsymbol{\mu} \end{array} \right.$$

Numerical Example - I

The investible universe consists of 4 risky assets. The risk return characteristics are as follows

Asset	μ	σ
A	0.04	0.07
B	0.08	0.12
C	0.12	0.18
D	0.15	0.26

Correlation Matrix $R = \begin{pmatrix} 1 & 0.2 & 0.5 & 0.3 \\ 0.2 & 1 & 0.7 & 0.4 \\ 0.5 & 0.7 & 1 & 0.9 \\ 0.3 & 0.4 & 0.9 & 1 \end{pmatrix}$

1. Calculate the covariance matrix Ω
2. Solve the following optimization problem

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \Omega \mathbf{w}$$

Subject to: $\mathbf{w}^T \boldsymbol{\mu} = 0.1$

$$\mathbf{w}^T \mathbf{1} = 1$$

3. Plot the risk return portfolio and identify the optimal portfolio.

Solution - I



1. The covariance matrix is conveniently calculated as

$$\mathbf{\Omega} = \mathbf{SRS} \quad \text{where}$$

$$\mathbf{S} = \text{diag}(\boldsymbol{\sigma}) = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & & & \\ 0 & \cdots & \cdots & \sigma_n \end{pmatrix} \quad \mathbf{R} = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{21} & 1 & \cdots & \rho_{2n} \\ \vdots & & & \\ \rho_{n1} & \cdots & \cdots & 1 \end{pmatrix}$$

Asset	μ	σ
A	0.04	0.07
B	0.08	0.12
C	0.12	0.18
D	0.15	0.26

$$\mathbf{R} = \begin{pmatrix} 1 & 0.2 & 0.5 & 0.3 \\ 0.2 & 1 & 0.7 & 0.4 \\ 0.5 & 0.7 & 1 & 0.9 \\ 0.3 & 0.4 & 0.9 & 1 \end{pmatrix} \quad \mathbf{S} = \begin{pmatrix} 0.07 & 0 & 0 & 0 \\ 0 & 0.12 & 0 & 0 \\ 0 & 0 & 0.18 & 0 \\ 0 & 0 & 0 & 0.26 \end{pmatrix} \quad \mathbf{\Sigma} = \begin{pmatrix} 0.0049 & 0.00168 & 0.0063 & 0.00546 \\ 0.00168 & 0.0144 & 0.01512 & 0.01248 \\ 0.0063 & 0.01512 & 0.0324 & 0.04212 \\ 0.00546 & 0.01248 & 0.04212 & 0.0676 \end{pmatrix}$$

Solution - I

2. The solution is given as

1 Calculate Σ^{-1}

$$\begin{pmatrix} 1301.39 & 1086.957 & -1909.36 & 883.8987 \\ 1086.957 & 1207.729 & -1952.5 & 905.7971 \\ -1909.36 & -1952.5 & 3390.589 & -1597.92 \\ 883.8987 & 905.7971 & -1597.92 & 771.8034 \end{pmatrix} \quad \text{unit} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

2 Calculate $A = \mathbf{1}^T \Sigma^{-1} \mathbf{1}$ $B = \boldsymbol{\mu}^T \Sigma^{-1} \mathbf{1}$ $C = \boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu}$

$$\begin{aligned} A &= 1505.261 \\ B &= 50.58862 \\ C &= 1.961295 \end{aligned}$$

3 Calculate $\lambda = \frac{Am - B}{AC - B^2}$ $\gamma = \frac{C - Bm}{AC - B^2}$ $m = 10\%$

$$\begin{aligned} \lambda &= 0.254261 \\ \gamma &= -0.00788 \end{aligned}$$

4 Calculate $\mathbf{w}^* = \Sigma^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1})$

$$\mathbf{w}^* = \begin{pmatrix} 0.058677 \\ 0.759027 \\ -0.31955 \\ 0.501841 \end{pmatrix}$$

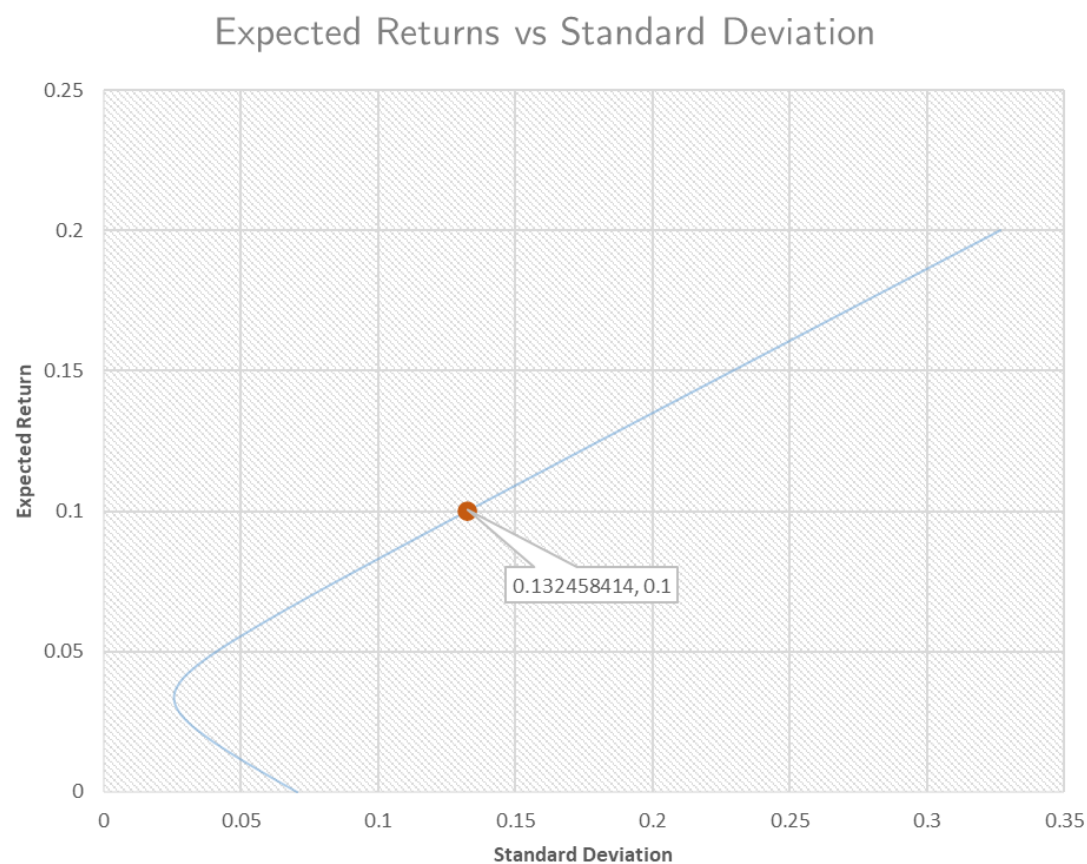
5 Calculate SD $\sigma = \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}$

$$\sigma[r_P] = 13.25\%$$

Solution - I



3. The boundary of the opportunity set (minimum variance frontier) can be found by solving the optimization problem at various levels of return threshold.





Global Minimum Variance Portfolio

- ▶ The minimum variance frontier parametrized by a return level (m) is also conveniently given as

$$\sigma_{\pi}^2(m) = \frac{Am^2 - 2Bm + C}{AC - B^2}$$

- ▶ The global minimum variance portfolio (GMVP) can be found by minimizing the above across all values of m . The objective is therefore

$$\min_m \sigma_{\pi}^2(m)$$

First order condition gives $\frac{d\sigma_{\pi}^2(m)}{dm} = 0 \Leftrightarrow \frac{2Am - 2B}{AC - B^2} \Leftrightarrow m = \frac{B}{A}$

The corresponding weight vector and portfolio variance is given as

$$w_G = \frac{\mathbf{\Omega}^{-1}\mathbf{1}}{A} \quad \sigma_G^2 = \frac{1}{A}$$

Numerical Example - II



- ▶ For the previous question, determine
 - ☐ The portfolio weights for the GMVP
 - ☐ The variance and return of GMVP

Solution - II



A	B	C
1505.261	50.58862	1.961295

$$\Sigma^{-1} = \begin{pmatrix} 1301.39 & 1086.957 & -1909.36 & 883.8987 \\ 1086.957 & 1207.729 & -1952.5 & 905.7971 \\ -1909.36 & -1952.5 & 3390.589 & -1597.92 \\ 883.8987 & 905.7971 & -1597.92 & 771.8034 \end{pmatrix}$$

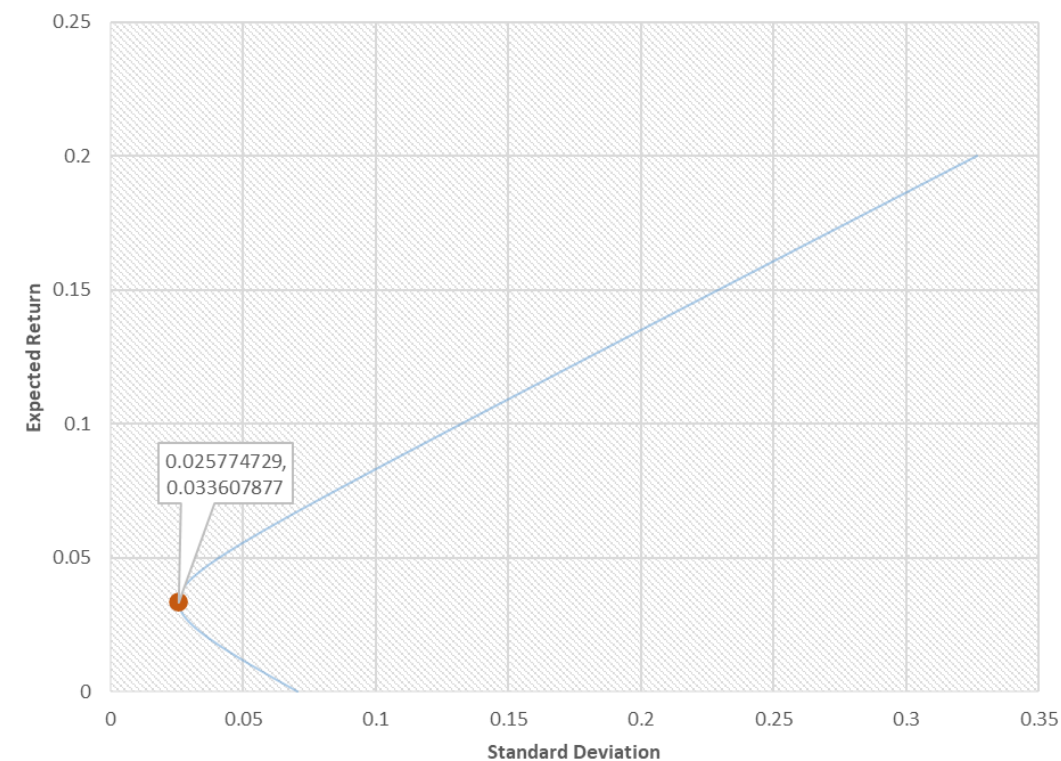
$$\mathbf{w}^* = \frac{1}{A} \mathbf{\Omega}^{-1} \mathbf{1}$$

$$\mathbf{w}^* = \begin{pmatrix} 0.905413 \\ 0.829084 \\ -1.37464 \\ 0.640142 \end{pmatrix}$$

$$\mathbb{E}[R_p] = \mathbf{w}^T \boldsymbol{\mu} = \frac{B}{A} = 3.361\%$$

$$\sigma[R_p] = \sqrt{\mathbf{w}^T \mathbf{\Omega} \mathbf{w}} = 2.577\%$$

Expected Returns vs Standard Deviation





With Risk Free Asset

- ▶ If our portfolio includes a risk-free asset, we can formulate the problem as follows. Let the weight of the risk-free asset be w_0 . We can frame our optimization based on weights of risky assets only i.e., $(w_1, w_2, \dots, w_n)$, and remove the weight constraint because any residual weight will be absorbed by w_0

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \mathbf{\Omega} \mathbf{w}$$

$$\text{Subject to: } r_f + \mathbf{w}^T (\boldsymbol{\mu} - r_f \mathbf{1}) = m$$

- ▶ The Lagrangian becomes $\mathcal{L}(\mathbf{w}, \lambda) = \frac{1}{2} \mathbf{w}^T \mathbf{\Omega} \mathbf{w} + \lambda (m - r_f - \mathbf{w}^T (\boldsymbol{\mu} - r_f \mathbf{1}))$

- ▶ Solving first order condition gives the optimal weights

$$\mathbf{w}^* = \frac{(m - r_f) \mathbf{\Omega}^{-1} (\boldsymbol{\mu} - r_f \mathbf{1})}{(\boldsymbol{\mu} - r_f \mathbf{1})^T \mathbf{\Omega}^{-1} (\boldsymbol{\mu} - r_f \mathbf{1})}$$



Tangency Portfolio

- ▶ The tangency portfolio is entirely invested in risky assets. The characteristics of the tangency portfolio are as follows :

Weight vector :
$$\mathbf{w}_t = \frac{\mathbf{\Omega}^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}{B - Ar_f}$$

Mean return :
$$m_t = \mathbf{w}_t^T \boldsymbol{\mu} = \frac{C - Br_f}{B - Ar_f}$$

SD:
$$\sigma_t = \sqrt{\mathbf{w}_t^T \mathbf{\Omega} \mathbf{w}_t} = \sqrt{\frac{C - 2Br_f + Ar_f^2}{(B - Ar_f)^2}}$$

Homework



1. Assuming risk-free return = 2.5%. What is the optimal allocation of the portfolio returning 10% ?
2. Find the allocations for tangency portfolio. What are the mean return and SD ?